

that draws 10 mA continuously? Assume the battery voltage is constant.

- 1-19 The maximum power a device can dissipate is 0.5 W. Determine the maximum current allowed by the device power rating when the voltage is 9 V.

- 1-20 A constant current of 2 A charges a battery for 4 hours. During the charging period the battery voltage is $v(t) = 12 - 2e^{-t}$, where t is in hours. Determine the energy stored in the battery.

- 1-21 Two electrical devices are connected as shown in Figure P1-21. Using the reference marks shown in the figure, find the power transferred and state whether the power is transferred from A to B or from B to A when

- (a) $v = +11$ V and $i = -1.1$ A
- (b) $v = +80$ V and $i = +20$ mA
- (c) $v = -120$ V and $i = -12$ mA
- (d) $v = -1.5$ V and $i = -600$ μ A

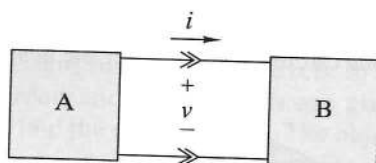


FIGURE P1-21

- 1-22 Figure P1-22 shows an electric circuit with a voltage and current variable assigned to each of the six devices. The device voltages and currents are observed to be

	v (V)	i (A)
Device 1	15	-1
Device 2	5	1
Device 3	10	2
Device 4	-10	-1
Device 5	20	-3
Device 6	20	2

Find the power associated with each device and state whether the device is absorbing or delivering power. Use the power balance to check your work.

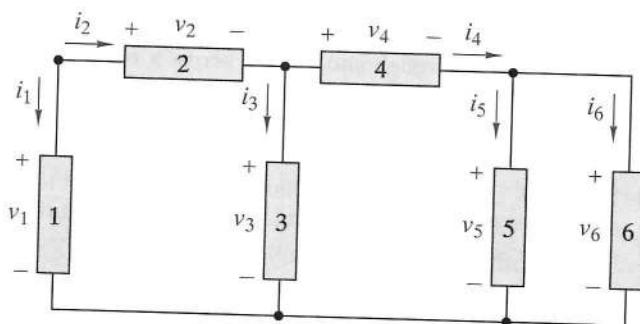


FIGURE P1-22

- 1-23 Figure P1-22 shows an electric circuit with a voltage and current variable assigned to each of the six devices. Use power balance to find v_4 when $v_1 = 20$ V, $i_1 = -2$ A, $p_2 = 20$ W, $p_3 = 10$ W, $i_4 = 1$ A, and $p_5 = p_6 = 2.5$ W. Is device 4 absorbing or delivering power?

- 1-24 Suppose that in Figure P1-22 a ground is connected to the minus (-) side of device 6 and another to the junction of devices 2, 3 and 4. Further, assume that the voltage v_4 is 5 V and v_1 is 10 V. What are the voltages v_2 , v_3 , v_5 , and v_6 ?

- 1-25 For $t \geq 0$ the voltage across and power absorbed by a two-terminal device are $v(t) = 2e^{-t}$ V and $p(t) = 40e^{-2t}$ mW. Find the total charge delivered to the device for $t \geq 0$.

OBJECTIVE 1-3 COMPUTATIONAL SOFTWARE INTRODUCTION (SECT. 1-4, APPENDIX B)

Given a simple computational problem use MATLAB as an appropriate engineering tool to solve the problem. Examples and exercises throughout the text, Appendix B

- 1-26 Repeat Problem 1-22 using MATLAB to perform the calculations. Create a vector for the voltage values, $v = [15 \ 5 \ 10 \ -10 \ 20 \ 20]$, and a vector for the current values, $i = [-1 \ 1 \ 2 \ -1 \ -3 \ 2]$. Compute the corresponding vector for the power values, p , using element-by-element multiplication ($.*$) and then use the sum command to verify the power balance.

- 1-27 Using the passive sign convention, the voltage across a device is $v(t) = 5 \cos(10t)$ V and the current through the device is $i(t) = 0.5 \sin(10t)$ A. Using MATLAB create a short script (m-file) to assign a value to the time variable, t , and then calculate the voltage, current, and power at that time. Run the script for $t = 0.2$ s and $t = 0.4$ s and for each result state whether the device is absorbing or delivering power.

INTEGRATING PROBLEMS

1-28 Power Ratio in dB

In complete darkness the voltage across and current through a two-terminal light detector are +6.5 V and +7 nA. In full sunlight the voltage and current are +1.1 V and +4.5 mA. Express the light/dark power ratio of the device in decibels (dB), where the power ratio in dB is

$$PR_{dB} \rightarrow 10 \log_{10}(p_2/p_1)$$

1-29 AC to DC Converter

A manufacturer's data sheet for the converter in Figure P1-29 states that the output voltage is $v_{dc} = 5$ V when the input voltage $v_{ac} = 120$ V. When the load draws a current $i_{dc} = 40$ A the input power is $p_{ac} = 300$ W. Find the efficiency of the converter.

SUMMARY

- An **electrical device** is a real physical entity, while a **circuit element** is a mathematical or graphical model that approximates major features of the device.
- Two-terminal circuit elements are represented by a circuit symbol and are characterized by a single constraint imposed on the associated current and voltage variables.
- An **electrical circuit** is an interconnection of electrical devices. The interconnections form nodes and loops.
- A **node** is an electrical juncture of the terminals of two or more devices. A **loop** is a closed path formed by tracing through a sequence of devices without passing through any node more than once.
- Device interconnections in a circuit lead to two connection constraints: **Kirchhoff's current law (KCL)** states that the algebraic sum of currents at a node is zero at every instant; and **Kirchhoff's voltage law (KVL)** states that the algebraic sum of voltages around any loop is zero at every instant.
- A pair of two-terminal elements are connected in **parallel** if they form a loop containing no other elements. The same voltage appears across any two elements connected in parallel.
- A pair of two-terminal elements are connected in **series** if they are connected at a node to which no other elements are connected. The same current exists in any two elements connected in series.
- Two circuits are said to be **equivalent** if they each have the same i - v constraints at a specified pair of terminals.
- Series and parallel equivalence and **voltage** and **current division** are important tools in circuit analysis and design.
- **Source transformation** changes a voltage source in series with a resistor into an equivalent current source in parallel with a resistor, or vice versa.
- **Circuit reduction** is a method of solving for selected signal variables in ladder circuits. The method involves sequential application of the series/parallel equivalence rules, source transformations, and the voltage/current division rules. The reduction sequence used depends on the variables to be determined and the structure of the circuit and is not unique.
- Computer-aided circuit analysis applies spreadsheets, circuit simulation, or math solver software to analyze circuit problems efficiently. The tools allow for visual solutions and can eliminate the need to perform tedious or lengthy manual calculations.

PROBLEMS

OBJECTIVE 2-1 ELEMENT CONSTRAINTS
(SECT. 2-1)

Given a two-terminal element with one or more electrical variables specified, use the element i - v constraint to find the magnitude and direction of the unknown variables. See Example 2-1

- 2-1 The current through a 33-k Ω resistor is 1.2 mA. Find the voltage across the resistor.
- 2-2 A 6.2-k Ω resistor dissipates 12 mW. Find the current through the resistor.
- 2-3 The conductance of a particular resistor is 1 mS. Find the current through the resistor when connected across a 9-V source.
- 2-4 In Figure P2-4 the resistor dissipates 25 mW. Find R_x .

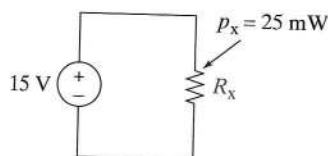


FIGURE P2-4

- 2-5 In Figure P2-5 find R_x and the power delivered to the resistor.

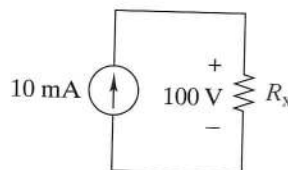


FIGURE P2-5

- 2-6 The i - v characteristic of a nonlinear resistor is $v = 75i + 0.2i^3$.
- (a) Calculate v and p for $i = \pm 0.5, \pm 1, \pm 2, \pm 5$, and ± 10 A.
- (b) Find the maximum error in v when the device is treated as a 75- Ω linear resistance on the range $|i| < 0.5$ A.
- 2-7 A 10-k Ω resistor has a power rating of $\frac{1}{8}$ W. Find the maximum voltage that can be applied to the resistor.
- 2-8 A certain type of film resistor is available with resistance values between 10 Ω and 100 M Ω . The maximum ratings for all resistors of this type are 500 V and $\frac{1}{4}$ W. Show that the voltage rating is the controlling limit for $R > 1$ M Ω and that the power rating is the controlling limit when $R < 1$ M Ω .

2-9 Figure P2-9 shows the circuit symbol for a class of two-terminal devices called diodes. The i - v relationship for a specific pn junction diode is $i = 2 \times 10^{-16}(e^{40v} - 1)$ A.

(a) Use this equation to find i and p for $v = 0, \pm 0.1, \pm 0.2, \pm 0.4$, and ± 0.8 V. Use these data to plot the i - v characteristic of the element.

(b) Is the diode linear or nonlinear, bilateral or nonbilateral, and active or passive?

(c) Use the diode model to predict i and p for $v = 5$ V. Do you think the model applies to voltages in this range? Explain.

(d) Repeat part (c) for $v = -5$ V.

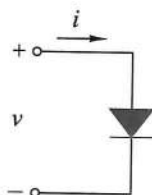


FIGURE P2-9

OBJECTIVE 2-2 CONNECTION CONSTRAINTS (SECT. 2-2)

Given a circuit composed of two-terminal elements:

- Identify nodes and loops in the circuit.
- Identify elements connected in series and in parallel.
- Use Kirchhoff's laws (KCL and KVL) to find selected signal variables.

See Examples 2-4, 2-5, 2-6 and Exercises 2-1, 2-2, 2-3, 2-4, 2-5

2-10 In Figure P2-10 $i_2 = -2$ A and $i_3 = 5$ A. Find i_1 and i_4

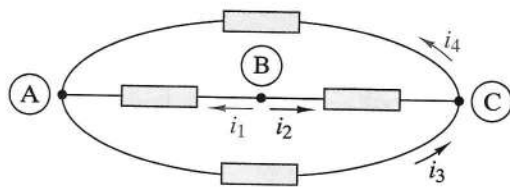


FIGURE P2-10

2-11 For the circuit in Figure P2-11:

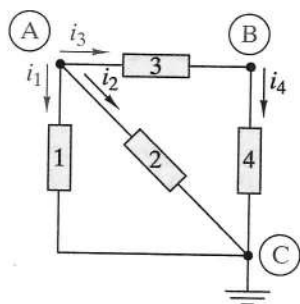


FIGURE P2-11

(a) Identify the nodes and at least two loops.

(b) Identify any elements connected in series or in parallel.

(c) Write KCL and KVL connection equations for the circuit.

2-12 In Figure P2-11 $i_2 = -10$ mA and $i_4 = 20$ mA. Find i_1 and i_3 .

2-13 For the circuit in Figure P2-13:

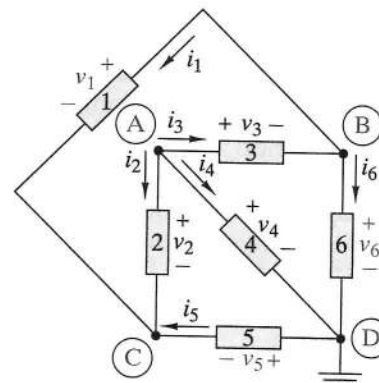


FIGURE P2-13

(a) Identify the nodes and at least three loops in the circuit.

(b) Identify any elements connected in series or in parallel.

(c) Write KCL and KVL connection equations for the circuit.

2-14 In Figure P2-13 $v_2 = 10$ V, $v_3 = -10$ V, and $v_4 = 3$ V. Find v_1 , v_5 , and v_6 .

2-15 The circuit in Figure P2-15 is organized around the three signal lines A, B, and C.

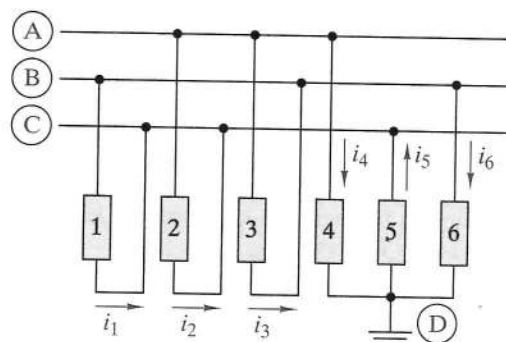


FIGURE P2-15

(a) Identify the nodes and at least three loops in the circuit.

(b) Write KCL connection equations for the circuit.

(c) If $i_1 = -20$ mA, $i_2 = -12$ mA, and $i_3 = 50$ mA, find i_4 , i_5 , and i_6 .

(d) Show that the circuit in Figure P2-15 is identical to that in Figure P2-13.

2-16 In Figure P2-16 $v_2 = 10$ V, $v_3 = 10$ V, and $v_4 = 10$ V. Find v_1 and v_5 .

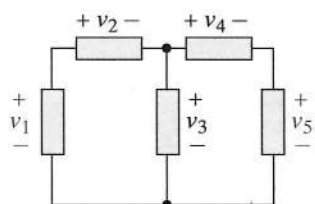


FIGURE P2-16

2-17 In Figure P2-17 $i_2 = 10 \text{ mA}$, $i_3 = -15 \text{ mA}$, and $i_4 = 5 \text{ mA}$. Find i_1 and i_5 .

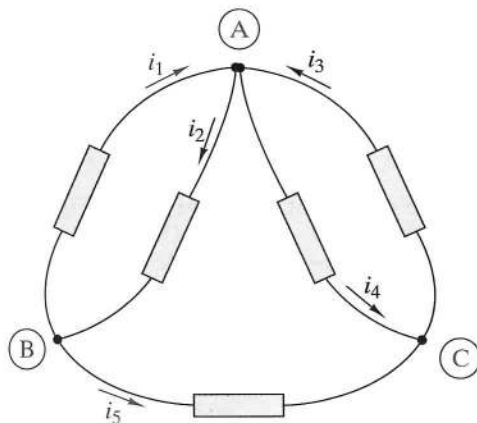


FIGURE P2-17

2-18 (a) Use the passive sign convention to assign voltage variables consistent with the currents in Figure P2-17. Write three KVL connection equations using these voltage variables.

(b) If $v_3 = 0 \text{ V}$, what can be said about the voltages across all the other elements?

2-19 The KCL equations for a three-node circuit are:

$$\text{Node A: } -i_1 + i_2 - i_4 = 0$$

$$\text{Node B: } -i_2 - i_3 + i_5 = 0$$

$$\text{Node C: } i_1 + i_3 + i_4 - i_5 = 0$$

Draw the circuit diagram and indicate the reference directions for the element currents.

OBJECTIVE 2-3 COMBINED CONSTRAINTS (SECT. 2-3)

Given a circuit consisting of independent sources and linear resistors, use the element constraints and connection constraints to find selected signal variables.

See Examples 2-7, 2-8, 2-9, 2-10 and Exercise 2-6

2-20 Find v_x and i_x in Figure P2-20.

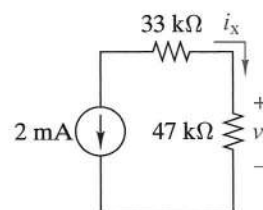


FIGURE P2-20

2-21 Find v_x and i_x in Figure P2-21.

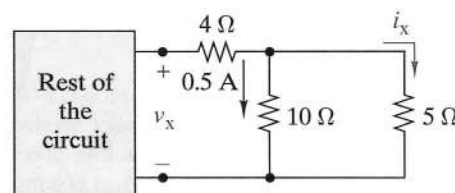


FIGURE P2-21

2-22 In Figure P2-22:

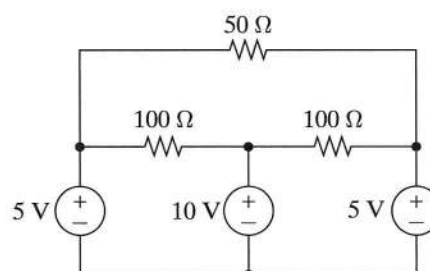


FIGURE P2-22

- Assign a voltage and current variable to every element.
- Use KVL to find the voltage across each resistor.
- Use Ohm's law to find the current through each resistor.
- Use KCL to find the current through each voltage source.

2-23 Find the power dissipated in the $1.5\text{-k}\Omega$ resistor in Figure P2-23.

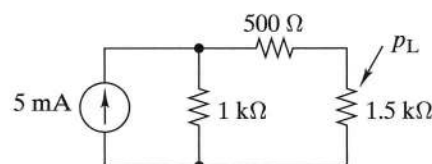


FIGURE P2-23

2-24 Figure P2-24 shows a subcircuit connected to the rest of the circuit at four points.

- Use element and connection constraints to find v_x and i_x .
- Show that the sum of the currents into the rest of the circuit is zero.

- (c) Find the voltage v_A with respect to the ground in the circuit.

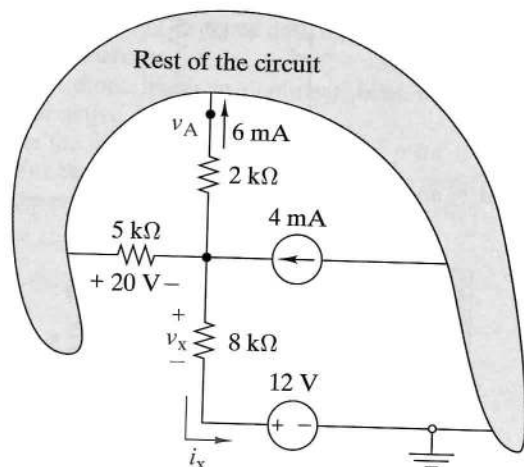


FIGURE P2-24

- 2-25 In Figure P2-25 $i_x = -0.5$ mA. Find the value of R .

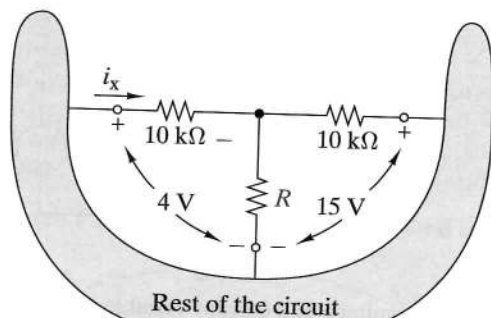


FIGURE P2-25

- 2-26 Figure P2-26 shows a resistor with one terminal connected to ground and the other connected to an arrow. The arrow symbol is used to indicate a connection to one terminal of a voltage source whose other terminal is connected to ground. The label next to the arrow indicates the source voltage at the ungrounded terminal. Find the voltage across, current through, and power dissipated in the resistor.

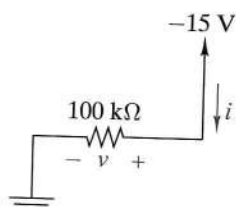


FIGURE P2-26

OBJECTIVE 2-4 EQUIVALENT CIRCUITS (SECT. 2-4)

- (a) Given a circuit consisting of linear resistors, find the equivalent resistance between a specified pair of terminals.
(b) Given a circuit consisting of a source-resistor combination find an equivalent source-resistor circuit.
See Examples 2-11, 2-12 and Exercises 2-7, 2-8, 2-9

- 2-27 Find the equivalent resistance R_{EQ} in Figure P2-27.

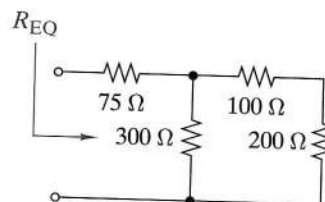


FIGURE P2-27

- 2-28 Find the equivalent resistance R_{EQ} in Figure P2-28.

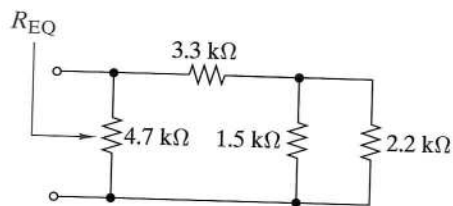


FIGURE P2-28

- 2-29 Find R_{EQ} in Figure P2-29 when the switch is open. Repeat when the switch is closed.

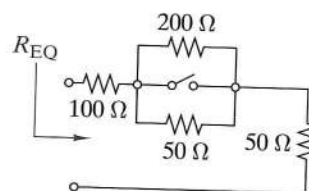


FIGURE P2-29

- 2-30 Show how the circuit in Figure P2-30 could be connected to achieve a resistance of 100 Ω, 200 Ω, 150 Ω, 50 Ω, 25 Ω, 33.3 Ω, and 133.3 Ω.

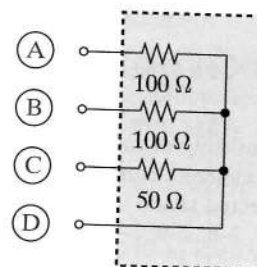


FIGURE P2-30

- 2-31 In Figure P2-31 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

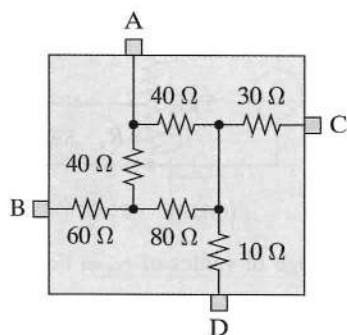


FIGURE P2-31

- 2-32 Select a value of R_L in Figure P2-32 so that $R_{EQ} = 25 \text{ k}\Omega$. Repeat for $R_{EQ} = 20 \text{ k}\Omega$.

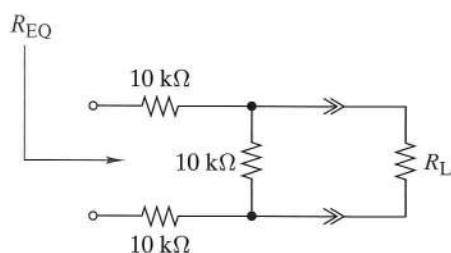


FIGURE P2-32

- 2-33 Using no more than four 1-kΩ resistors, show how the following equivalent resistors can be constructed: 2 kΩ, 500 Ω, 1.5 kΩ, 333 Ω, 250 Ω, and 400 Ω.

- 2-34 Find the equivalent practical voltage source at terminals A and B in Figure P2-34.

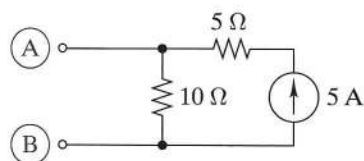


FIGURE P2-34

- 2-35 In Figure P2-35 the $i-v$ characteristic of network N is $v + 50i = 5 \text{ V}$. Find the equivalent practical current source for the network.

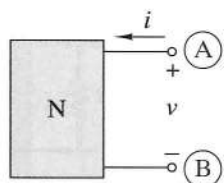


FIGURE P2-35

- 2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60 \text{ k}\Omega$.

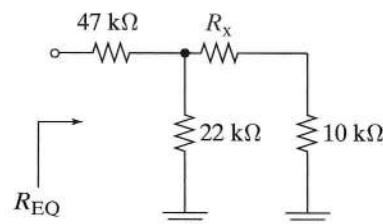


FIGURE P2-36

- 2-37 Two 10-kΩ potentiometers (a potentiometer is a variable resistor whose value between the two ends is 10 kΩ and between one end and the wiper—the third terminal—can range from 0 Ω to 10 kΩ) are connected as shown in Figure P2-37. What is the range of R_{EQ} ?

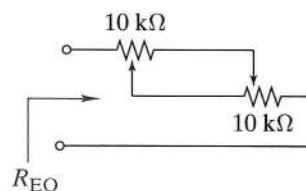


FIGURE P2-37

- 2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

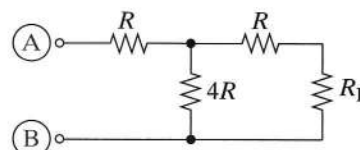


FIGURE P2-38

- 2-39 What is the range of R_{EQ} in Figure P2-39?

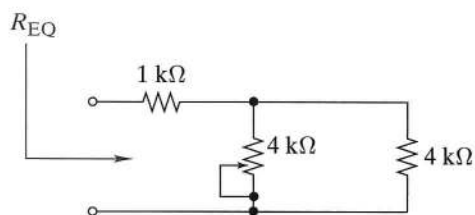


FIGURE P2-39

- 2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.